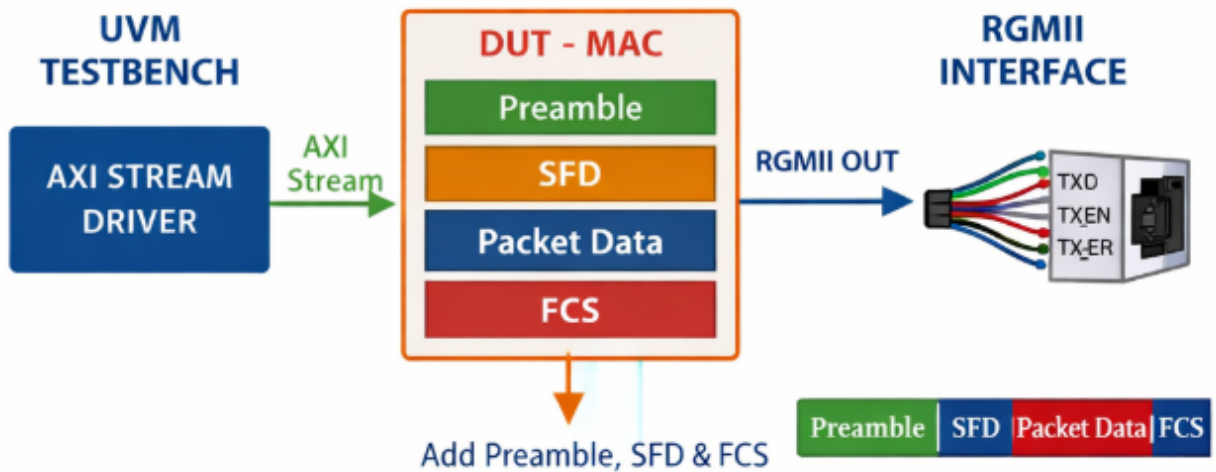
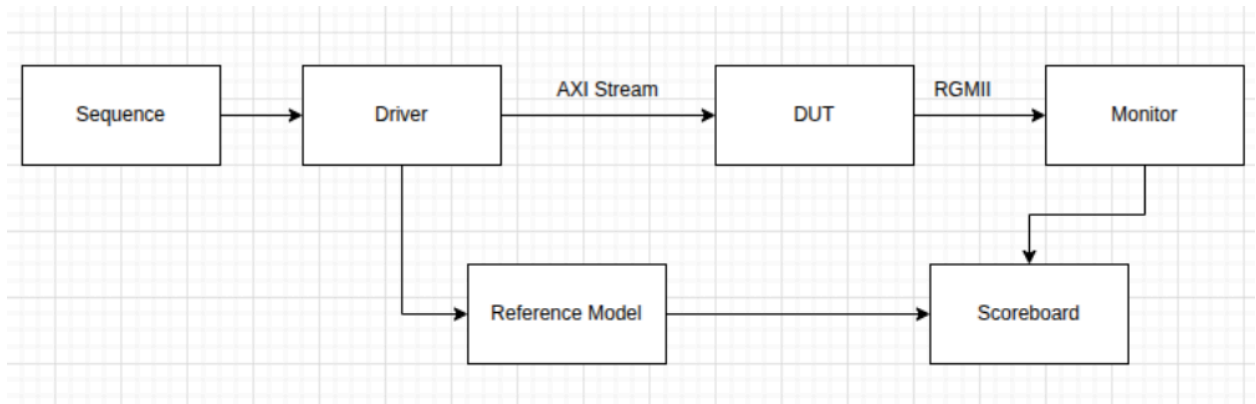


Eth TX Path



Flow:

- We send packets to DUT through axi-s interface
- Format of what we send for tx path:
 $\text{dest_addr}(6\text{B}) + \text{src_addr}(6\text{B}) + \text{eth_type}(2\text{B}) + \text{payload}(46\text{B}-1500\text{B})$
- DUT(MAC) adds preamble, SFD and FCS/CRC and then the packet becomes:
 $\text{Preamble}(7\text{B}) + \text{SFD}(1\text{B}) + \text{dest_addr}(6\text{B}) + \text{src_addr}(6\text{B}) + \text{eth_type}(2\text{B})$
 $+ \text{payload}(46\text{B}-1500\text{B}) + \text{FCS}(4\text{B})$
- DUT sends this out on RGMII.
- Verif Block Diagram:



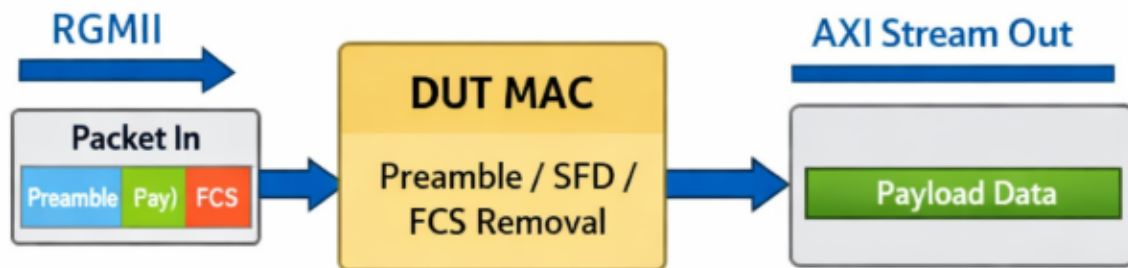
Observations:

- Design is not reading tsrtrb correctly. In last flit/beat when tlast is asserted, tsrtrb indicates valid bytes and DUT should consider only those bytes and discard the rest. For example if tsrtrb is 8'b00000111 it means only three bytes(LSB) are valid but DUT considers all 8 bytes valid.
- According to specs, zeros should be pad if payld is less or equal to 46B, But design accepts min 64B frame, so 14 bytes of dest, src and eth_type, + 50Bytes of payld makes

64B frame. So, min payld should be 50B and padding must be applicable for payld less than 50B, not 46B.

- Cannot send 1518B frame due to dependency on tstrb bug, we have to send 8 Byte aligned frames till the bug is fixed and 1518 is not 8B aligned, so had to send 1512B packet and passed

Eth RX Path



- RGMII Input
- MAC: Remove Preamble, SFD, FCS
- AXI Stream Output

Flow:

- We send packets to DUT through RGMII.
- We send to DUT:
Preamble(7B) + SFD(1B) + dest_addr(6B) + src_addr(6B) + eth_type(2B)
+ payload(46B-1500B) + FCS(4B)
- MAC removes preamble, sfd and crc and packet becomes:
dest_addr(6B) + src_addr(6B) + eth_type(2B) + payload(46B-1500B)
- This is sent out on the axi-stream interface.

Formation of Packet:

- Preamble: Standard, 7 byte 55555555555555
- SFD: Standard, 1 byte D5
- Destination MAC address: 6 Byte random
- Source MAC Address: 6 Byte random
- Eth_Type: 2 Byte, depending on next header
- Payload: Random
- CRC: 4 Byte, calculation procedure is added in the general understanding section.

General Understanding

- Preamble's standard value is 7 bytes of (55555555555555).
- SFD's standard value is 1 byte D5.

- CRC is calculated on dest_addr, src_addr, eth_type and payload.
- CRC Process (Logic implemented in reference model):
 1. Initialization: The 32-bit crc variable is initialized to all ones (32'hFFFFFFFF).
 2. Byte Processing: The function iterates through every byte in the data array one by one.
 3. XOR Input: The current data byte is XORed directly into the crc register.
 4. Bit-Level Iteration: A nested loop runs 8 times per byte, processing the data one bit at a time.
 5. Condition Check: The function checks the least significant bit (crc[0]):
 - a. If bit 0 is 1: The register is shifted right by one bit and then XORed with the reflected polynomial 32'hEDB88320.
 - b. If bit 0 is 0: The register is simply shifted right by one bit.
 6. Completion: After all bytes and their 8 bits are processed, the function performs a bitwise NOT (~crc) to invert the result.
 7. Output: The final 32-bit inverted value is returned as the Frame Check Sequence (FCS).
- In promiscuous mode, all the frames are accepted. This mode is enabled when register config1[16] = 1. This mode is used in the rx path.
- AXI Stream Protocol:
 1. Initialization: The system begins in a reset state (typically active-low ARESETN), where TVALID must be driven low to ensure no accidental data transfer.
 2. Handshake Mechanism: Data transfer occurs only on the rising edge of the clock when both the master and slave are ready:
 - a. Master Ready (TVALID): The master asserts TVALID to indicate it has valid data on the TDATA bus.
 - b. Slave Ready (TREADY): The slave asserts TREADY to indicate it can accept data in the current clock cycle.
 3. Handshake Completion: A "beat" (single unit of data) is successfully transferred when TVALID and TREADY are both high at the same time.
 4. Packet Boundaries (TLAST): To indicate the end of a sequence or frame (like the last pixel in a video frame), the master asserts the TLAST signal alongside the final data beat.
 5. Optional Signaling: Depending on the design, additional signals like TKEEP (valid bytes) or TUSER (user-defined metadata, like "Start of Frame") may be used to provide more context for the stream
 6. Tstrb indicates valid Bytes in a beat. If a bit in TSTRB is low, the corresponding byte on TDATA is ignored by the receiver, but it still counts toward the total byte count of the packet.
- MAC is working in little endian, let's understand little endian:
Imagine you have a 32-bit hexadecimal number: 0x12345678. How it is broken down:
 - 12: Most Significant Byte (MSB) — the "big" end.
 - 34: Middle byte.
 - 56: Middle byte.
 - 78: Least Significant Byte (LSB) — the "little" end.

If you start storing this value at memory address 1000, a Little-Endian system lays it out like this:

Address	Value	Significance
1000	78	LSB
1001	56	Middle
1002	34	Middle
1003	12	MSB

- Unused Signals:

phy_int_n	PHY interrupt (not used, tied high)
phy_pme_n	PHY power management (not used)
phy_mdio_i	MDIO input (tied low)
phy_mdio_o	MDIO output (unconnected)
phy_mdio_oe	MDIO output enable (unconnected)
phy_mdc	MDIO clock (unconnected)
tx_axis_tid_i	No description given and no significance observed

- In 10Mbps and 100Mbps, RGMII must work in sdr(single data rate) mode

5.0 10/100 Functionality

This interface can be used to implement the 10/100 Mbps Ethernet Media Independent Interface (MII) by reducing the clock rate to 25MHz for 100Mbps operation and 2.5MHz for 10Mbps. The TXC will always be generated by the MAC and RXC will be generated by the PHY. During packet reception, the RXC may be stretched on either the positive or negative pulse to accommodate the transition from the free running clock to a data-synchronous clock domain. When the speed of the PHY changes, a similar stretching of the positive or negative pulses is allowed. No glitching of the clocks are allowed during speed transitions.

This interface will operate at 10 and 100Mbps speeds exactly the same way it does at Gigabit speed with the exception that the data may be duplicated on the ↓ edge of the appropriate clock.

The MAC will hold TX_CTL low until it has ensured that it is operating at the same speed as the PHY.

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